Research Progress Report

Statistical Methods and Literature Review for Various Scales and Application in Mental Health

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1. Introduction

Statistical analysis and literature review constitute two fundamental components of scientific research, providing the methodological framework for experimental design, data analysis, and evidence-based interpretation of research findings. A sound understanding of statistical methodologies, together with a critical evaluation of existing literature, is essential for developing scientifically rigorous and reproducible research.

This report presents the theoretical and methodological studies undertaken in preparation for research on machine learning applications in mental healthcare. The statistical component covers both parametric and non-parametric hypothesis testing techniques, including one-sample and two-sample hypothesis tests, one-way and two-way Analysis of Variance (ANOVA), statistical power analysis, and sample size estimation. Emphasis is placed on the mathematical formulations, underlying assumptions, and practical applicability of these statistical methods in research.

The report further examines the practical application of statistical methodologies through the analysis of psychometric scale development and validation studies involving SERATS, SERAPTS, and FEATS. These studies demonstrate the application of reliability analysis, validity assessment, factor analysis, and other statistical procedures used in the development and evaluation of psychological measurement instruments.

In addition, the report reviews contemporary machine learning research in mental healthcare through the analysis of a systematic review and representative research papers. Particular emphasis is placed on understanding the objectives, data sources, machine learning methodologies, and research contributions of the reviewed studies, together with the identification of current research trends, limitations, and future research opportunities.

Collectively, the studies presented in this report establish the statistical and methodological foundation required for the proposed research. They provide the necessary background for understanding existing research methodologies, selecting appropriate analytical techniques, and defining the direction of future work in machine learning applications for mental healthcare.

2. Study of Statistical Methods

2.1 Introduction

Statistical methods form the foundation of quantitative research by providing systematic techniques for hypothesis testing, parameter estimation, and experimental design. Selecting an appropriate statistical procedure requires an understanding of its mathematical formulation, assumptions, and applicability to the problem under investigation.

This chapter presents the statistical methods studied during the reporting period. The mathematical foundations, assumptions, statistical models, test statistics, power analysis, and sample size estimation procedures are discussed to provide the theoretical basis for the research undertaken in this work.

2.2 Parametric Statistical Methods

Parametric statistical methods were studied to understand the principles of statistical inference for data that satisfy assumptions such as normality, independence, and homogeneity of variance. These methods form the basis of quantitative data analysis and are widely used for testing hypotheses and comparing population parameters.

2.2.1 One-Sample Hypothesis Testing

One-sample hypothesis testing is a parametric statistical procedure used to determine whether the mean of a sample differs significantly from a known or hypothesized population mean. The choice of statistical test depends primarily on whether the population standard deviation is known. When the population variance is known and the sample size is sufficiently large, the one-sample Z-test is employed. When the population variance is unknown, the one-sample t-test is used, which accounts for the additional uncertainty introduced by estimating the population variance from the sample.

Hypothesis Formulation

The objective of one-sample hypothesis testing is to determine whether the population mean differs significantly from a specified (hypothesized) population mean. Depending on the research objective, the hypotheses may be formulated as shown in Table 2.1.

Table 2.1: Hypotheses for One-Sample Hypothesis Testing

Test Type	**Null Hypothesis**	**Alternative Hypothesis**
Two-tailed	$H_0 : \mu = \mu_0$	$H_1 : \mu \neq \mu_0$
Right-tailed	$H_0 : \mu = \mu_0$	$H_1 : \mu > \mu_0$
Left-tailed	$H_0 : \mu = \mu_0$	$H_1 : \mu < \mu_0$

The notation used in Table 2.1 is summarized in Table 2.2.

Table 2.2: Notation Used in One-Sample Hypothesis Testing

Symbol	**Description**
μ	Population mean
μ_0	Hypothesized population mean
H_0	Null hypothesis
H_1	Alternative hypothesis

Assumptions

The following assumptions are considered for one-sample hypothesis testing:

1. The observations are randomly selected.
2. The observations are independent.
3. The data are measured on an interval or ratio scale.
4. The population is approximately normally distributed.
5. For the Z-test, the population standard deviation (σ) is known.
6. For the t-test, the population standard deviation is unknown and is estimated using the sample standard deviation.

One-Sample Z-Test

The one-sample Z-test is applicable when the population standard deviation (σ) is known and the observations are drawn from a normally distributed population. Under these conditions, the sampling distribution of the sample mean follows a standard normal distribution.

The test statistic is given by

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}},$$

where

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is the sample mean,

$$\sigma$$

is the known population standard deviation, and

$$n$$

denotes the sample size.

Under the null hypothesis,

$$Z \sim N(0,1),$$

where $N(0,1)$ represents the standard normal distribution.

The null hypothesis is rejected if the calculated Z-statistic lies within the critical region corresponding to the chosen significance level α, or equivalently if

$$p < \alpha.$$

One-Sample t-Test

The one-sample t-test is employed when the population standard deviation is unknown. In this case, the population variance is estimated using the sample variance,

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

The corresponding test statistic is

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}},$$

where

$$S = \sqrt{S^2}$$

is the sample standard deviation.

Under the null hypothesis, the test statistic follows Student's t-distribution with

$$df = n - 1$$

degrees of freedom.

The null hypothesis is rejected if the calculated t-statistic lies within the critical region of the t-distribution or, equivalently, if

$$p < \alpha.$$

Comparison between Z-test and t-test

The Z-test is appropriate when the population standard deviation is known, whereas the t-test is preferred when the population standard deviation is unknown. As the sample size increases, the t-distribution approaches the standard normal distribution, making the results of both tests increasingly similar.

Applications

One-sample hypothesis testing is widely applied in quality control, biomedical research, psychology, education, and social sciences to determine whether a sample differs significantly from a known or hypothesized population value.

2.2.2 Two-Sample Hypothesis Testing

Two-sample hypothesis testing is a parametric statistical method used to determine whether two independent or dependent populations differ significantly with respect to their population means. The selection of an appropriate statistical test depends on the availability of population variances, the equality of variances between the two populations, and whether the observations are independent or paired. Accordingly, four different cases of two-sample hypothesis testing were studied during the reporting period.

Hypothesis Formulation

The objective of two-sample hypothesis testing is to determine whether the means of two populations differ significantly. Depending on the research objective, the hypotheses may be formulated as shown in Table 2.3.

Table 2.3: Hypotheses for Two-Sample Hypothesis Testing

Test Type	**Null Hypothesis**	**Alternative Hypothesis**
Two-tailed	$H_0 : \mu_1 = \mu_2$	$H_1 : \mu_1 \neq \mu_2$
Right-tailed	$H_0 : \mu_1 = \mu_2$	$H_1 : \mu_1 > \mu_2$
Left-tailed	$H_0 : \mu_1 = \mu_2$	$H_1 : \mu_1 < \mu_2$

The notation used in Table 2.3 is summarized in Table 2.4.

Table 2.4: Notation Used in Two-Sample Hypothesis Testing

Symbol	**Description**
μ_1	Population mean of the first population
μ_2	Population mean of the second population
H_0	Null hypothesis
H_1	Alternative hypothesis

Assumptions

The choice of the appropriate two-sample test depends on the following assumptions.

1. Whether the population variances are known or unknown.
2. If the variances are unknown, whether they can be assumed equal or unequal.
3. Whether the two samples are independent or paired (dependent).
4. The populations are approximately normally distributed and observations are randomly selected.

Based on these assumptions, four statistical models are available for comparing two population means.

Case I: Two-Sample Z-Test

The two-sample Z-test is applicable when the population standard deviations of both populations are known and the two samples are independent. Under these conditions, the difference between the sample means follows a standard normal distribution.

The test statistic is given by

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

where

Symbol	Description
$\bar{X}, \bar{Y}$	Sample means of the two groups
μ_1, μ_2	Population means
σ_1, σ_2	Known population standard deviations
n_1, n_2	Sample sizes

Under the null hypothesis,

$$Z \sim N(0, 1),$$

where $N(0, 1)$ denotes the standard normal distribution.

The null hypothesis is rejected if the calculated Z-statistic falls within the critical region or equivalently if

$$p < \alpha.$$

Case II: Pooled Two-Sample t-Test

The pooled two-sample t-test is employed when the population variances are unknown but can reasonably be assumed equal. A pooled estimate of the common variance is first obtained and then used to construct the test statistic.

The pooled variance is

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

The corresponding test statistic is

$$t = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

where

Symbol	Description
S_p	Pooled sample standard deviation
S_1, S_2	Sample standard deviations
n_1, n_2	Sample sizes
$\bar{X}, \bar{Y}$	Sample means

Under the null hypothesis, the test statistic follows Student's t-distribution with

$$df = n_1 + n_2 - 2.$$

The null hypothesis is rejected if

$$p < \alpha.$$

Case III: Welch's t-Test

Welch's t-test is applicable when the population variances are unknown and cannot be assumed equal. Unlike the pooled t-test, this method does not require the assumption of equal variances.

The test statistic is

$$t = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}.$$

The degrees of freedom are approximated using the Welch–Satterthwaite equation,

$$df = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}}.$$

where

Symbol	Description
$\bar{X}, \bar{Y}$	Sample means
S_1, S_2	Sample standard deviations
n_1, n_2	Sample sizes
df	Approximate degrees of freedom

The null hypothesis is rejected if

$$p < \alpha.$$

Case IV: Paired t-Test

The paired t-test is used when observations are collected as matched pairs or repeated measurements on the same experimental units. Rather than comparing two independent sample means, the analysis is performed on the paired differences.

For each pair,

$$D_i = X_i - Y_i,$$

and the mean difference is

$$\bar{D} = \frac{1}{n} \sum_{i=1}^{n} D_i.$$

The test statistic is

$$t = \frac{\bar{D}}{S_D/\sqrt{n}},$$

where

Symbol	Description
$\bar{D}$	Mean of paired differences
S_D	Standard deviation of paired differences
n	Number of pairs

Under the null hypothesis, the test statistic follows Student's t-distribution with

$$df = n - 1.$$

The null hypothesis is rejected if

$$p < \alpha.$$

Decision Rule

For each of the above tests, the calculated test statistic is compared with the corresponding critical value at the chosen significance level α. Alternatively, the null hypothesis is rejected if

$$p < \alpha.$$

Applications

Two-sample hypothesis testing is widely applied in medicine, psychology, education, engineering, agriculture, and social sciences for comparing the means of two populations under different experimental conditions.

2.2.3 One-Way Analysis of Variance (ANOVA)

One-way Analysis of Variance (ANOVA) is a parametric statistical technique used to determine whether there are statistically significant differences among the means of three or more independent groups. Unlike multiple two-sample hypothesis tests, one-way ANOVA simultaneously compares all group means while controlling the overall Type I error rate. The method is based on partitioning the total variability observed in the data into variation due to differences between groups and variation within groups.

Mathematical Model

Suppose there are k independent groups with sample sizes $n_1, n_2, \ldots, n_k$.

The observation corresponding to the jth subject in the ith group is represented as

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij},$$

where

Table 2.5: Notation Used in One-Way ANOVA

Symbol	Description
Y_{ij}	Observation from the jth subject in the ith group
μ	Overall (grand) mean
τ_i	Effect of the ith treatment/group
ε_{ij}	Random error term
k	Number of groups
n_i	Sample size of the ith group

The random errors are assumed to be independently and normally distributed,

$$\varepsilon_{ij} \sim N(0, \sigma^2).$$

Hypothesis Formulation

The objective of one-way ANOVA is to determine whether all population means are equal.

The hypotheses are summarized in Table 2.6.

Table 2.6: Hypotheses for One-Way ANOVA

Hypothesis	Expression
Null Hypothesis	$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$
Alternative Hypothesis	H_1 : At least one population mean differs.

Assumptions

The validity of one-way ANOVA depends on the following assumptions.

1. The observations are randomly selected.
2. The observations are independent.
3. The populations are normally distributed.
4. The population variances are equal (homogeneity of variance).

Partition of Variability

One-way ANOVA partitions the total variation into two components,

$$SS_T = SS_B + SS_W,$$

where

$$SS_T$$

is the Total Sum of Squares,

$$SS_B$$

is the Between-Group Sum of Squares, and

$$SS_W$$

is the Within-Group Sum of Squares.
These quantities are computed as

$$SS_T = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y})^2,$$

$$SS_B = \sum_{i=1}^{k} n_i (\bar{Y}_i - \bar{Y})^2,$$

$$SS_W = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2.$$

ANOVA Table

The analysis of variance is summarized using the ANOVA table shown below.

Table 2.7: Structure of One-Way ANOVA Table

Source	SS	df	MS	F
Between Groups	SS_B	$k-1$	$\frac{SS_B}{k-1}$	$\frac{MS_B}{MS_W}$
Within Groups	SS_W	$N-k$	$\frac{SS_W}{N-k}$	–
Total	SS_T	$N-1$	–	–

F-Test Statistic

The F-statistic used in one-way ANOVA is

$$F = \frac{MS_B}{MS_W},$$

where

$$MS_B = \frac{SS_B}{k-1},$$

and

$$MS_W = \frac{SS_W}{N-k}.$$

Under the null hypothesis,

$$F \sim F(k-1, N-k).$$

Decision Rule

The null hypothesis is rejected if the calculated F-statistic exceeds the critical value obtained from the F-distribution for the chosen significance level α, or equivalently if

$$p < \alpha.$$

Applications

One-way ANOVA is widely used in agriculture, medicine, psychology, education, engineering, and the social sciences to compare the means of three or more independent groups and determine whether significant differences exist among them.

2.2.4 Two-Way Analysis of Variance (ANOVA)

Two-way Analysis of Variance (ANOVA) is an extension of one-way ANOVA that is used to evaluate the effects of two independent factors on a response variable simultaneously. In addition to examining the individual (main) effects of each factor, two-way ANOVA determines whether there is a significant interaction between the two factors. This approach provides a more comprehensive analysis of experimental data while reducing the number of separate statistical tests required.

Mathematical Model

Suppose Factor A has a levels and Factor B has b levels. Let Y_{ijk} denote the observation corresponding to the kth replication under the ith level of Factor A and the jth level of Factor B.

The two-way ANOVA model is

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk},$$

where

Table 2.8: Notation Used in Two-Way ANOVA

Symbol	**Description**
Y_{ijk}	Observation corresponding to the kth replication
μ	Overall (grand) mean
α_i	Effect of the ith level of Factor A
β_j	Effect of the jth level of Factor B
$(\alpha\beta)_{ij}$	Interaction effect between Factors A and B
ε_{ijk}	Random error term

The random errors are assumed to be independently and normally distributed,

$$\varepsilon_{ijk} \sim N(0, \sigma^2).$$

Hypothesis Formulation

Two-way ANOVA simultaneously tests the significance of the main effects of Factors A and B, together with their interaction effect.

The hypotheses are summarized in Table 2.9.

Table 2.9: Hypotheses for Two-Way ANOVA

Effect	**Null Hypothesis**	**Alternative Hypothesis**
Factor A	$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_a = 0$	At least one $\alpha_i \neq 0$
Factor B	$H_0 : \beta_1 = \beta_2 = \cdots = \beta_b = 0$	At least one $\beta_j \neq 0$
Interaction	$H_0 : (\alpha\beta)_{ij} = 0$	At least one interaction effect exists

Assumptions

The validity of two-way ANOVA depends on the following assumptions.

1. The observations are randomly selected.
2. The observations are independent.
3. The populations are normally distributed.
4. The population variances are equal (homogeneity of variance).

Partition of Variability

In two-way ANOVA, the total variation is partitioned into variation due to Factor A, Factor B, their interaction, and random error.

$$SS_T = SS_A + SS_B + SS_{AB} + SS_E.$$

where

- SS_A : Sum of Squares due to Factor A.
- SS_B : Sum of Squares due to Factor B.
- SS_{AB} : Sum of Squares due to interaction.
- SS_E : Error Sum of Squares.

ANOVA Table

Table 2.10: Structure of Two-Way ANOVA Table

Source	SS	df	MS	F
Factor A	SS_A	$a-1$	$\frac{SS_A}{a-1}$	$\frac{MS_A}{MS_E}$
Factor B	SS_B	$b-1$	$\frac{SS_B}{b-1}$	$\frac{MS_B}{MS_E}$
Interaction	SS_{AB}	$(a-1)(b-1)$	$\frac{SS_{AB}}{(a-1)(b-1)}$	$\frac{MS_{AB}}{MS_E}$
Error	SS_E	$N-ab$	$\frac{SS_E}{N-ab}$	–
Total	SS_T	$N-1$	–	–

F-Test Statistics

The F-statistics for testing the significance of the main effects and interaction effect are

$$F_A = \frac{MS_A}{MS_E},$$

$$F_B = \frac{MS_B}{MS_E},$$

and

$$F_{AB} = \frac{MS_{AB}}{MS_E}.$$

Each statistic follows an F-distribution under its corresponding null hypothesis.

Decision Rule

The null hypothesis corresponding to each effect is rejected if the calculated F-statistic exceeds the critical value from the F-distribution at the chosen significance level α, or equivalently if

$$p < \alpha.$$

Applications

Two-way ANOVA is widely used in medicine, agriculture, engineering, psychology, education, and industrial experiments where the effects of two independent factors and their interaction on a response variable are of interest.

2.3 Statistical Power and Sample Size Estimation

Statistical power analysis is an essential component of experimental design and hypothesis testing. It provides a quantitative framework for determining the probability of detecting a true effect while controlling the risks of incorrect statistical decisions. Power analysis is closely related to the concepts of Type I error, Type II error, effect size, significance level, and sample size. Together, these quantities assist researchers in designing experiments that are both statistically reliable and scientifically meaningful.

2.3.1 Hypothesis Testing Errors

Every hypothesis test may lead to one of two incorrect decisions.

A Type I error occurs when the null hypothesis is rejected even though it is true.

A Type II error occurs when the null hypothesis is not rejected even though the alternative hypothesis is true.

These probabilities are denoted by

$$\alpha = P(\text{Reject } H_0 | H_0 \text{ true})$$

and

$$\beta = P(\text{Fail to Reject } H_0 | H_1 \text{ true}).$$

Table 2.11: Possible Outcomes of Hypothesis Testing

Actual Situation	Decision	Outcome
H_0 true	Reject H_0	Type I Error (α)
H_0 true	Fail to Reject H_0	Correct Decision
H_1 true	Reject H_0	Correct Decision (Power)
H_1 true	Fail to Reject H_0	Type II Error (β)

2.3.2 Power Functions of Statistical Tests

The power of a statistical test depends on the probability distribution of the test statistic under the alternative hypothesis. Since different hypothesis tests have different rejection regions, the corresponding power functions also differ. The following subsections present the power functions for the hypothesis tests studied during this work.

Power Function of the One-Sample Z-Test

Consider a random sample drawn from a normal population with known variance. The standardized non-centrality parameter is

$$\lambda = \frac{\mu - \mu_0}{\sigma/\sqrt{n}},$$

where

Symbol	Description
μ	True population mean
μ_0	Hypothesized mean
σ	Population standard deviation
n	Sample size
λ	Non-centrality parameter

For the hypotheses

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu > \mu_0,$$

the rejection region is

$$Z > Z_\alpha.$$

The corresponding power function is

$$\boxed{Power = 1 - \Phi\left(Z_\alpha - \lambda\right)}$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

For

$$H_1 : \mu < \mu_0,$$

the rejection region becomes

$$Z < -Z_\alpha,$$

and the power function is

$$\boxed{Power = \Phi\left(-Z_\alpha - \lambda\right)}$$

For

$$H_1 : \mu \neq \mu_0,$$

the rejection region is

$$|Z| > Z_{\alpha/2}.$$

The corresponding power function is

$$\boxed{Power = \Phi\left(-Z_{\alpha/2} - \lambda\right) + 1 - \Phi\left(Z_{\alpha/2} - \lambda\right)}$$

Power Function of the One-Sample t-Test

When the population variance is unknown, the one-sample t-test is employed. Under the alternative hypothesis, the test statistic follows a non-central Student's t-distribution. The corresponding non-centrality parameter is

$$\delta = \frac{\mu - \mu_0}{\sigma/\sqrt{n}},$$

where μ is the true population mean, μ_0 is the hypothesized mean, σ is the population standard deviation, and n is the sample size.

The statistical power is defined as

$$Power = P\left(T > t_\alpha \mid \delta\right)$$

for a one-tailed test, and

$$Power = P\left(|T| > t_{\alpha/2} \mid \delta\right)$$

for a two-tailed test.

For sufficiently large sample sizes, the non-central t-distribution can be approximated by the standard normal distribution. The corresponding power functions become

$$Power = 1 - \Phi\left(t_\alpha - \delta\right)$$

for a one-tailed test, and

$$Power = \Phi\left(-t_{\alpha/2} - \delta\right) + 1 - \Phi\left(t_{\alpha/2} - \delta\right)$$

for a two-tailed test,

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

Power Function of Two-Sample Tests

For two independent samples with equal population variances, the non-centrality parameter is

$$\delta = \frac{\mu_1 - \mu_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

where μ_1 and μ_2 denote the population means, σ is the common population standard deviation, and n_1 and n_2 are the respective sample sizes.

The statistical power is defined as

$$Power = P\left(T > t_\alpha \mid \delta\right)$$

for a one-tailed test, and

$$Power = P\left(|T| > t_{\alpha/2} \mid \delta\right)$$

for a two-tailed test.

Using the normal approximation for large samples, the power functions become

$$Power = 1 - \Phi\left(Z_\alpha - \delta\right)$$

for a one-tailed test, and

$$Power = \Phi\left(-Z_{\alpha/2} - \delta\right) + 1 - \Phi\left(Z_{\alpha/2} - \delta\right)$$

for a two-tailed test.

Power Function of ANOVA

The statistical power of the ANOVA F-test is determined using the non-central F-distribution. The corresponding non-centrality parameter is

$$\lambda = \frac{\sum_{i=1}^{k} n_i(\mu_i - \mu)^2}{\sigma^2},$$

where k is the number of groups, n_i is the sample size of the ith group, μ_i is the mean of the ith group, μ is the overall mean, and σ^2 is the common population variance.

The statistical power is given by

$$Power = P\left(F > F_\alpha \mid \lambda\right),$$

where the test statistic follows a non-central F-distribution under the alternative hypothesis.

The corresponding cumulative distribution function may be written as

$$Power = 1 - F_F\left(F_\alpha; df_1, df_2, \lambda\right),$$

where $F_F(\cdot)$ denotes the cumulative distribution function of the non-central F-distribution, while df_1 and df_2 represent the numerator and denominator degrees of freedom, respectively.

2.3.3 Effect Size

Effect size measures the magnitude of the difference or relationship being investigated. Unlike the p-value, which only indicates statistical significance, the effect size quantifies the practical importance of the observed effect.

For comparing two population means, Cohen's standardized effect size is defined as

$$d = \frac{\mu_1 - \mu_2}{\sigma},$$

where

Table 2.12: Notation Used in Effect Size

Symbol	**Description**
d	Cohen's effect size
μ_1, μ_2	Population means
σ	Common population standard deviation

According to Cohen's guideline,

Table 2.13: Interpretation of Cohen's Effect Size

Effect Size	Interpretation
0.20	Small
0.50	Medium
0.80	Large

2.3.4 Relationship Between Statistical Power and Sample Size

Statistical power is influenced by several factors, including the significance level, sample size, effect size, and variability of the data. These factors are interrelated and must be considered together while designing an experiment.

The relationships among these quantities are summarized in Table 2.14.

Table 2.14: Factors Affecting Statistical Power

Factor	**Change**	**Effect on Power**
Sample Size	Increase	Increase
Effect Size	Increase	Increase
Significance Level (α)	Increase	Increase
Population Variability	Increase	Decrease

Consequently, increasing the sample size or effect size generally increases the probability of detecting true differences, whereas larger population variability reduces statistical power.

2.3.5 Sample Size Estimation

Sample size estimation is performed before conducting an experiment to ensure that the study possesses sufficient statistical power while avoiding unnecessary data collection. The required sample size depends on the significance level, desired statistical power, population variability, and expected effect size.

For estimating a population mean, the required sample size is

$$n = \left(\frac{(Z_{1-\alpha/2} + Z_{1-\beta})\sigma}{\delta} \right)^2,$$

where

Table 2.15: Notation Used in Sample Size Estimation

Symbol	**Description**
n	Required sample size
$Z_{1-\alpha/2}$	Standard normal critical value
$Z_{1-\beta}$	Standard normal value corresponding to desired power
σ	Population standard deviation
δ	Minimum detectable difference

The estimated sample size should always be rounded upward to ensure that the desired statistical power is achieved.

2.3.6 Applications

Statistical power analysis and sample size estimation are widely applied in medicine, psychology, education, engineering, agriculture, and social science research. These techniques assist researchers in designing experiments that are capable of detecting meaningful effects while minimizing the risks of Type I and Type II errors. Proper power analysis contributes to reliable statistical inference and improves the overall quality and reproducibility of scientific studies.

2.4 Non-Parametric Statistical Methods

Non-parametric statistical methods are distribution-free techniques that do not require strict assumptions regarding the underlying population distribution. These methods are particularly useful when the assumptions of parametric tests, such as normality or homogeneity of variance, are violated. Instead of relying on the actual numerical values of the observations, non-parametric tests generally operate on the signs or ranks of the data, making them more robust to outliers and skewed distributions.

During the reporting period, three commonly used non-parametric hypothesis tests were studied: the One-Sample Sign Test, the Wilcoxon Signed-Rank Test, and the Wilcoxon Rank-Sum Test. Their mathematical formulations, assumptions, test statistics, power analysis, and sample size estimation procedures were examined.

2.4.1 One-Sample Sign Test

The one-sample sign test is a non-parametric alternative to the one-sample t-test. It is used to determine whether the median of a population differs from a specified value without assuming any particular population distribution.

Hypothesis Formulation

The hypotheses are summarized in Table 2.16.

Table 2.16: Hypotheses for the One-Sample Sign Test

Test Type	**Null Hypothesis**	**Alternative Hypothesis**
Two-tailed	$H_0 : M = M_0$	$H_1 : M \neq M_0$
Right-tailed	$H_0 : M = M_0$	$H_1 : M > M_0$
Left-tailed	$H_0 : M = M_0$	$H_1 : M < M_0$

Assumptions

The one-sample sign test is based on the following assumptions.

1. Observations are randomly selected.
2. Observations are independent.
3. The response variable is measured on at least an ordinal scale.
4. The population distribution need not be normal.

Test Statistic

Each observation is compared with the hypothesized median M_0. Positive and negative differences are recorded, while observations equal to M_0 are discarded.

If S denotes the number of positive signs, then

$$S \sim Bin(n, 0.5)$$

under the null hypothesis.

For sufficiently large sample sizes,

$$Z = \frac{S - \frac{n}{2}}{\sqrt{\frac{n}{4}}}$$

approximately follows the standard normal distribution.

Decision Rule

The null hypothesis is rejected if the calculated test statistic falls within the critical region or, equivalently, if

$$p < \alpha.$$

Power Function

If the probability of obtaining a positive sign under the alternative hypothesis is denoted by p, then the power of the one-sample sign test is

$$Power = P(S \geq c \mid p) = \sum_{x=c}^{n} \binom{n}{x} p^x (1-p)^{n-x},$$

where c is the critical value determined by the significance level α.

Sample Size Estimation

Using the normal approximation to the binomial distribution, the required sample size may be approximated by

$$n = \frac{\left(Z_{1-\alpha}\sqrt{p_0(1-p_0)} + Z_{1-\beta}\sqrt{p_1(1-p_1)}\right)^2}{(p_1 - p_0)^2},$$

where

- p_0 denotes the probability under the null hypothesis,
- p_1 denotes the anticipated probability under the alternative hypothesis.

2.4.2 Wilcoxon Rank-Sum Test

The Wilcoxon Rank-Sum Test is a non-parametric statistical procedure used to compare two independent populations when the assumptions of the independent two-sample t-test are not satisfied. Instead of comparing sample means, the test evaluates whether the two populations have identical distributions by ranking all observations jointly. Because it is based on ranks rather than the original observations, the test is robust against non-normality and outliers.

Hypothesis Formulation

The objective of the Wilcoxon Rank-Sum Test is to determine whether two independent populations have the same central tendency.

The hypotheses are summarized in Table 2.17.

Table 2.17: Hypotheses for the Wilcoxon Rank-Sum Test

Test Type	**Null Hypothesis**	**Alternative Hypothesis**
Two-tailed	$H_0 : M_1 = M_2$	$H_1 : M_1 \neq M_2$
Right-tailed	$H_0 : M_1 = M_2$	$H_1 : M_1 > M_2$
Left-tailed	$H_0 : M_1 = M_2$	$H_1 : M_1 < M_2$

Assumptions

The Wilcoxon Rank-Sum Test is based on the following assumptions.

1. The two samples are randomly selected.
2. The two samples are independent.

3. The observations are measured on at least an ordinal scale.

4. The observations can be ranked.

5. The two populations have similar distributional shapes.

Test Statistic

Suppose the two independent samples have sizes n_1 and n_2. All observations are combined and ranked in ascending order. Let

$$R_1$$

denote the sum of the ranks corresponding to the first sample.
The Wilcoxon Rank-Sum statistic is

$$W = R_1.$$

An equivalent statistic is the Mann–Whitney statistic,

$$U = R_1 - \frac{n_1(n_1+1)}{2}.$$

For sufficiently large sample sizes, the standardized test statistic is

$$Z = \frac{U - \mu_U}{\sigma_U},$$

where

$$\mu_U = \frac{n_1 n_2}{2},$$

and

$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}.$$

Under the null hypothesis,

$$Z \sim N(0,1).$$

Decision Rule

The null hypothesis is rejected if the calculated test statistic lies within the critical region corresponding to the chosen significance level α, or equivalently if

$$p < \alpha.$$

Power Function

The exact statistical power of the Wilcoxon Rank-Sum Test is obtained from the sampling distribution of the rank-sum statistic under the alternative hypothesis. Let

$$W$$

denote the rank-sum statistic. The statistical power is defined as

$$Power = P\left(W \in C \mid H_1\right),$$

where C represents the rejection region.

For moderate and large sample sizes, the normal approximation yields

$$Power = 1 - \Phi\left(Z_\alpha - \delta\right)$$

for a one-tailed test, and

$$Power = \Phi\left(-Z_{\alpha/2} - \delta\right) + 1 - \Phi\left(Z_{\alpha/2} - \delta\right)$$

for a two-tailed test, where δ denotes the standardized effect size.

Sample Size Estimation

Using the asymptotic normal approximation, the required sample size for each group is estimated by

$$n = 2\left(\frac{Z_{1-\alpha} + Z_{1-\beta}}{\delta}\right)^2$$

for a one-tailed test, and

$$n = 2\left(\frac{Z_{1-\alpha/2} + Z_{1-\beta}}{\delta}\right)^2$$

for a two-tailed test,

where δ denotes the standardized effect size.

In practice, exact sample size determination is generally performed using iterative numerical procedures or dedicated statistical software because it depends on the distribution of the rank statistic.

Applications

The Wilcoxon Rank-Sum Test is widely applied in medicine, psychology, biology, engineering, and the social sciences for comparing two independent groups when the

assumptions of the independent two-sample t-test are violated. It is particularly useful for analysing ordinal data and continuous data that are not normally distributed.

2.4.3 Wilcoxon Signed-Rank Test

The Wilcoxon Signed-Rank Test is a non-parametric statistical procedure used to compare two related or paired samples when the assumptions of the paired t-test are not satisfied. Unlike the Sign Test, which considers only the direction of the differences, the Wilcoxon Signed-Rank Test incorporates both the direction and the magnitude of the paired differences by ranking their absolute values. Consequently, it generally provides greater statistical power while remaining free from the assumption of normality.

Hypothesis Formulation

The objective of the Wilcoxon Signed-Rank Test is to determine whether the median difference between two paired populations is zero. The hypotheses are summarized in Table 2.18.

Table 2.18: Hypotheses for the Wilcoxon Signed-Rank Test

Test Type	**Null Hypothesis**	**Alternative Hypothesis**
Two-tailed	$H_0 : M_D = 0$	$H_1 : M_D \neq 0$
Right-tailed	$H_0 : M_D = 0$	$H_1 : M_D > 0$
Left-tailed	$H_0 : M_D = 0$	$H_1 : M_D < 0$

where M_D denotes the population median of the paired differences.

Assumptions

The Wilcoxon Signed-Rank Test is based on the following assumptions.

1. The observations are collected as matched pairs.
2. Each pair is selected independently of the others.
3. The paired differences are measured on at least an ordinal scale.
4. The distribution of the paired differences is symmetric about the median.
5. Zero differences are omitted before assigning ranks.

Test Statistic

Let

$$D_i = X_i - Y_i, \qquad i = 1, 2, \ldots, n,$$

denote the paired differences, where X_i and Y_i represent the observations in the two related samples.

Observations with

$$D_i = 0$$

are discarded. The remaining absolute differences

$$|D_i|$$

are ranked from the smallest to the largest, and each rank is assigned the sign of its corresponding difference.

The sums of the positive and negative ranks are

$$W^+ = \sum R_i^+,$$

and

$$W^- = \sum R_i^-,$$

respectively.

The Wilcoxon Signed-Rank statistic is

$$T = \min(W^+, W^-).$$

For sufficiently large sample sizes, the standardized test statistic is

$$Z = \frac{T - \mu_T}{\sigma_T},$$

where

$$\mu_T = \frac{n(n+1)}{4},$$

and

$$\sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{24}}.$$

Under the null hypothesis,

$$Z \sim N(0,1).$$

Decision Rule

The null hypothesis is rejected if the calculated test statistic lies within the critical region corresponding to the chosen significance level α, or equivalently if

$$p < \alpha.$$

Power Function

The exact statistical power of the Wilcoxon Signed-Rank Test is obtained from the sampling distribution of the signed-rank statistic under the alternative hypothesis.

Let

$$T$$

denote the Wilcoxon Signed-Rank statistic. The statistical power is defined as

$$Power = P\left(T \in C \mid H_1\right),$$

where C denotes the rejection region.

For moderate and large sample sizes, the standardized statistic follows an approximate normal distribution. Consequently, the power function for a one-tailed test is

$$Power = 1 - \Phi\left(Z_\alpha - \delta\right),$$

while for a two-tailed test it is

$$Power = \Phi\left(-Z_{\alpha/2} - \delta\right) + 1 - \Phi\left(Z_{\alpha/2} - \delta\right),$$

where δ denotes the standardized effect size and $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution.

Sample Size Estimation

Using the asymptotic normal approximation, the required sample size is estimated by

$$n = \left(\frac{Z_{1-\alpha} + Z_{1-\beta}}{\delta}\right)^2$$

for a one-tailed test, and

$$n = \left(\frac{Z_{1-\alpha/2} + Z_{1-\beta}}{\delta} \right)^2$$

for a two-tailed test,

where δ represents the standardized effect size.

In practice, exact sample size calculations are generally obtained using iterative numerical methods or dedicated statistical software because they depend on the distribution of the signed-rank statistic.

Applications

The Wilcoxon Signed-Rank Test is widely used in medicine, psychology, education, biology, and social science research for analysing paired or matched observations. It is particularly useful for before-and-after studies, repeated measurements on the same subjects, and experiments involving ordinal or non-normally distributed continuous data.

2.5 Mathematical Comparison of Parametric and Non-Parametric Methods

The study also included a comparative analysis of the mathematical formulations of parametric and non-parametric statistical methods. The comparison focused on the corresponding test statistics, statistical power functions, and sample size estimation procedures. This analysis provides insight into how non-parametric methods replace distribution-based statistics with sign- or rank-based statistics while preserving the objectives of hypothesis testing.

Table 2.19: Comparison of Test Statistics

Method	**Parametric Form**	**Non-Parametric Form**
One Sample	$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$S \sim Bin(n, 0.5)$ (Sign Test)
Paired Samples	$t = \frac{\bar{D}}{S_D/\sqrt{n}}$	$T = \min(W^+, W^-)$ (Wilcoxon Signed-Rank)
Independent Samples	$t = \frac{\bar{X}_1 - \bar{X}_2}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$	$U = R_1 - \frac{n_1(n_1+1)}{2}$ (Wilcoxon Rank-Sum)

Table 2.20: Comparison of Statistical Power Functions

Method	**Parametric**	**Non-Parametric**
One Sample	$Power = 1 - \Phi(Z_\alpha - \delta)$	$Power = P(S \in C \mid H_1)$
Paired Samples	$Power = P(T > t_\alpha \mid \delta)$	$Power = P(W \in C \mid H_1)$
Independent Samples	$Power = P(T > t_\alpha \mid \delta)$	$Power = P(U \in C \mid H_1)$

Table 2.21: Comparison of Sample Size Estimation

Method	**Parametric**	**Non-Parametric**
One Sample	$n = \left(\frac{Z_{1-\alpha}+Z_{1-\beta}}{\delta}\right)^2$	Normal approximation based on the binomial distribution
Paired Samples	$n = \left(\frac{Z_{1-\alpha}+Z_{1-\beta}}{\delta}\right)^2$	Normal approximation based on the signed-rank statistic
Independent Samples	$n = 2\left(\frac{Z_{1-\alpha}+Z_{1-\beta}}{\delta}\right)^2$	Normal approximation based on the rank-sum statistic

2.6 Summary

This chapter presented the statistical methods studied during the reporting period, including both parametric and non-parametric approaches to hypothesis testing. The mathematical formulations, assumptions, test statistics, statistical power functions, and sample size estimation procedures for the studied methods were discussed in detail. A comparative analysis between the corresponding parametric and non-parametric techniques was also carried out to highlight their similarities, differences, and applicability. The concepts presented in this chapter establish the statistical foundation required for the subsequent literature review and future research in machine learning applications for mental health.

3. Literature Review

3.1 Introduction

A comprehensive review of the existing literature is an essential step in the research process, as it provides an understanding of current methodologies, identifies research gaps, and supports the development of a well-founded research direction. In addition to establishing theoretical knowledge, literature review enables researchers to evaluate existing statistical techniques, machine learning approaches, and experimental methodologies employed in related studies.

During the reporting period, an extensive literature survey was conducted in the areas of psychometric scale validation and machine learning applications in mental health. The study included the analysis of psychometric validation studies, a systematic review of machine learning techniques for mental health, and several original research articles referenced within the review. The knowledge gained through these studies, together with presentations and discussions with the research guide and assistant guide, contributed significantly to the development of the proposed research direction.

This chapter presents the literature reviewed during the reporting period, summarizes the key findings from the selected studies, and highlights the knowledge and research insights acquired in preparation for the proposed work.

3.2 Psychometric Scale Development and Validation Studies

Psychometric scales are standardized measurement instruments designed to assess psychological constructs that cannot be measured directly, such as emotions, behaviours, personality traits, and mental health conditions. Before these instruments can be applied in research or clinical practice, they must undergo rigorous psychometric evaluation to establish their reliability, validity, and ability to accurately measure the intended constructs. Such validation ensures that the resulting measurements are scientifically meaningful, reproducible, and suitable for monitoring therapeutic outcomes.

During the reporting period, three psychometric studies related to art therapy and mental health were reviewed to understand the statistical procedures involved in scale development and validation. The studies included the development of the Self-Expression and Emotion Regulation in Art Therapy Scale (SERATS), the extension and validation of the Self-Expression and Emotion Regulation in Arts and Psychomotor Therapies Scale (SERAPTS), and the validation of the Problem-Solving Scale of the Formal Elements Art Therapy Scale (FEATS). Together, these studies demonstrated different approaches to psychometric evaluation, including questionnaire construction, factor analysis, reliability assessment, validity testing, sensitivity to change, and criterion-based validation.

The analysis of these studies provided practical insight into how statistical methods are applied in psychological measurement, enabling a deeper understanding of the processes involved in developing and evaluating reliable assessment instruments for clinical and research applications.

During the reporting period, a detailed study of psychometric scale validation procedures was undertaken through the analysis of three validation studies: the Self-Expression and Emotion Regulation in Art Therapy Scale (SERATS), the Self-Expression and Emotion Regulation in Art Therapy for Parent–Child Therapy Scale (SERAPTS), and the Family Expressive Responses in Art Therapy Scale (FERATS). These studies provided practical insight into the statistical techniques employed for assessing the reliability, validity, and overall quality of psychological measurement instruments.

3.2.1 Self-Expression and Emotion Regulation in Art Therapy Scale (SERATS)

Background

Although art therapy has been widely incorporated into multidisciplinary treatment programmes for individuals with personality disorders, few instruments have been specifically designed to evaluate its therapeutic outcomes. Most existing psychological questionnaires assess general mental health or behavioural symptoms rather than the unique therapeutic effects of art therapy. To address this limitation, Haeyen *et al.* developed the Self-Expression and Emotion Regulation in Art Therapy Scale (SERATS), a psychometric instrument specifically designed to measure self-expression and emotion regulation achieved through art therapy.

Objective

The study aimed to develop a concise psychometric instrument capable of objectively evaluating the contribution of art therapy within multidisciplinary treatment programmes.

The authors further sought to establish the reliability, validity, and responsiveness of the scale so that it could be employed both as an outcome measure and as a routine monitoring instrument during therapy.

Methodology

The questionnaire was developed from qualitative interviews and expert consultation, resulting in an initial pool of 26 items describing the perceived effects of art therapy. The instrument was subsequently administered to patients diagnosed with Cluster B and Cluster C personality disorders receiving multidisciplinary treatment that included art therapy. The validation process consisted of sequential psychometric analyses to identify the optimal questionnaire structure and evaluate its measurement properties.

Statistical Validation

The validation process consisted of several complementary statistical analyses, each serving a specific purpose in establishing the psychometric quality of the proposed instrument.

Table 3.1: Statistical Validation Procedure of SERATS

Statistical Analysis	**Purpose in the Study**
Exploratory Factor Analysis	To identify the underlying latent structure of the questionnaire and reduce the initial item pool to the most representative items.
Confirmatory Factor Analysis	To verify that the identified factor structure adequately represented the observed responses.
Cronbach's Alpha	To determine whether the questionnaire items consistently measured the same psychological construct.
Test–Retest Reliability	To evaluate the stability of the questionnaire scores over repeated measurements.
Construct Validity	To determine whether SERATS measured a construct distinct from existing psychological questionnaires.
Sensitivity to Change	To examine whether the scale was capable of detecting therapeutic improvement throughout treatment.

Major Findings

The statistical analyses resulted in a unidimensional nine-item questionnaire representing self-expression and emotion regulation during art therapy. Exploratory and confirmatory factor analyses supported a single-factor structure, while the scale demonstrated excellent internal consistency (Cronbach's $\alpha = 0.94$) and very high test–retest reliability ($r = 0.96$). Construct validity analyses indicated that SERATS measured a distinct psychological construct rather than general mental health symptoms, and sensitivity analyses confirmed its ability to monitor therapeutic change over time. These findings established SERATS as a reliable and clinically useful outcome measure for art therapy.

Learning Outcomes

This study provided practical understanding of the complete process involved in psychometric scale development, beginning with questionnaire construction and concluding with statistical validation. It demonstrated how different statistical techniques collectively establish the reliability, validity, and clinical applicability of a psychological measurement

instrument.

3.2.2 Self-Expression and Emotion Regulation in Arts and Psychomotor Therapies Scale (SERAPTS)

Background

Although SERATS demonstrated strong psychometric properties for visual art therapy, no equivalent monitoring instrument existed for other creative arts and psychomotor therapies. Consequently, the questionnaire was extended and adapted into the Self-Expression and Emotion Regulation in Arts and Psychomotor Therapies Scale (SERAPTS) to support outcome monitoring across visual art, dance, drama, music, and psychomotor therapies.

Objective

The objective of the study was to evaluate the reliability and sensitivity to change of the adapted SERAPTS questionnaire when applied to multiple forms of creative arts and psychomotor therapies. The study further investigated whether the adapted instrument could serve as a common monitoring tool across different therapeutic modalities.

Methodology

A pre–post study design was conducted involving adult patients diagnosed with emotion regulation problems or Cluster B/C personality disorders receiving arts and psychomotor therapies. Participants completed the SERAPTS questionnaire at baseline and again after three months of treatment. Internal consistency was evaluated using Cronbach's alpha, while paired-sample t-tests and Cohen's effect size were employed to assess sensitivity to therapeutic change.

Statistical Validation

Unlike the original SERATS study, the primary objective of this work was not to develop a new instrument but to determine whether the existing questionnaire could be reliably applied across different forms of creative arts and psychomotor therapies. Consequently, the statistical analyses focused on evaluating the reliability of the adapted instrument and its ability to detect therapeutic changes over time.

Table 3.2: Statistical Validation Procedure of SERAPTS

Statistical Analysis	**Purpose in the Study**
Cronbach's Alpha	To evaluate the internal consistency of the questionnaire across different therapy modalities.
Paired Sample t-Test	To determine whether statistically significant therapeutic improvement occurred during treatment.
Sensitivity to Change	To evaluate the capability of SERAPTS to monitor patient progress over time.
Cohen's Effect Size	To quantify the magnitude of the observed therapeutic improvement.

Major Findings

The adapted questionnaire demonstrated excellent reliability across all therapy modalities, with an overall Cronbach's α of 0.95. Individual therapy forms also exhibited high internal consistency, including dance therapy ($\alpha = 0.88$), drama therapy ($\alpha = 0.89$), music therapy ($\alpha = 0.95$), and psychomotor therapy ($\alpha = 0.95$). Furthermore, a statistically significant improvement was observed after three months of therapy ($t = -4.39$, $p < 0.001$), corresponding to a medium effect size (Cohen's $d = 0.54$). These findings demonstrated that SERAPTS can reliably monitor treatment outcomes across multiple creative arts and psychomotor therapies.

Learning Outcomes

The SERAPTS study demonstrated how an existing psychometric instrument can be extended to new therapeutic domains while preserving its reliability and measurement quality. The study also illustrated the practical application of internal consistency analysis, paired-sample hypothesis testing, and effect size estimation for evaluating longitudinal treatment outcomes.

3.2.3 Formal Elements of Art Therapy Scale (FEATS)

Background

The Formal Elements of Art Therapy Scale (FEATS) is one of the most widely used assessment instruments in art therapy. Originally developed for evaluating the Person Picking an Apple from a Tree (PPAT) drawing task in adults, FEATS consists of

fourteen rating scales that assess different formal characteristics of drawings. Among these, the Problem-Solving Scale evaluates how effectively an individual represents the process of solving the task of picking an apple from a tree. Since children's cognitive development differs substantially from adults, further validation was required before applying this scale to younger populations.

Objective

The primary objective of the study was to further validate the FEATS Problem-Solving Scale for preschool children aged 5–6.5 years. Specifically, the authors investigated whether the scores obtained from the FEATS Problem-Solving Scale were associated with children's executive functions and intrinsic motivation. In addition, a complementary assessment tool, namely the Children's Picking Process Scale (CPPS), was developed to evaluate how children represented the picking process within their drawings.

Statistical Validation

Unlike the previous studies, this work focused on criterion-based validation by examining whether the FEATS Problem-Solving Scale was associated with established measures of executive functioning and motivation. The statistical analyses employed in the study are summarized in Table 3.3.

Table 3.3: Statistical Validation Procedure of FEATS

Statistical Analysis	**Purpose in the Study**
Inter-rater Reliability (Cronbach's Alpha)	To evaluate the consistency between independent raters scoring the PPAT drawings.
Internal Consistency	To assess the reliability of the BRIEF executive function scales and the Children's Motivation Scale.
Spearman Correlation	To examine the relationship between FEATS, CPPS, executive functions, and motivation.
Pearson Correlation	To evaluate the relationships among the executive function subscales.
Pairwise Comparison	To compare executive function scores across different CPPS categories.

Major Findings

The FEATS Problem-Solving Scale demonstrated good inter-rater reliability (Cronbach's $\alpha = 0.85$), while the newly proposed CPPS exhibited excellent reliability (Cronbach's $\alpha = 0.985$). A strong positive correlation was observed between FEATS and CPPS ($\rho = 0.742, p < 0.01$), supporting the validity of the newly proposed scale. Furthermore, significant relationships were identified between FEATS scores and two executive function domains, namely planning and organising and initiation, indicating that children's problem-solving representations were associated with their cognitive abilities. In addition, children's motivation showed a significant positive correlation with FEATS scores but not with CPPS, suggesting that successful problem solving in drawings reflects both executive functioning and goal-directed behaviour.

Learning Outcomes

This study demonstrated an alternative approach to psychometric validation by evaluating an existing assessment instrument against external behavioural and cognitive measures. Unlike SERATS and SERAPTS, which primarily focused on questionnaire development and reliability assessment, FEATS illustrated how criterion-related validation can be established through correlation analyses with independent psychological constructs. The study also provided practical understanding of inter-rater reliability, correlation analysis, and the interpretation of validation evidence within psychological assessment research.

3.3 Systematic Review on Machine Learning in Mental Health

3.3.1 Introduction

Machine learning has become an important research area in mental healthcare due to its ability to analyse large volumes of heterogeneous data for supporting early diagnosis, symptom monitoring, prognosis, and personalized interventions. Recent advances in sensing technologies, digital health records, social media platforms, and mobile devices have further expanded the availability of data that can be utilized for developing intelligent mental health support systems. However, despite the rapid growth of research in this domain, the practical implementation of machine learning models in clinical settings remains a significant challenge because of issues related to data quality, model interpretability, privacy, and human–computer interaction.

To obtain a comprehensive understanding of the current state of research, an in-depth study was conducted on the systematic review entitled *Machine Learning in*

Mental Health: A Systematic Review of the HCI Literature to Support the Development of Effective and Implementable ML Systems. The review analyzed existing research from both machine learning and Human–Computer Interaction (HCI) perspectives, providing valuable insights into the datasets, machine learning techniques, application domains, implementation challenges, and future research opportunities within mental healthcare.

Unlike a conventional literature survey that primarily summarizes the findings of the review paper, the present study focused on analysing the individual research articles discussed within the systematic review. Particular emphasis was placed on understanding the objectives of each study, the data sources employed, the machine learning models adopted, the contributions made by the researchers, and the limitations identified in the existing literature. This detailed analysis provided a deeper understanding of the current research landscape and assisted in identifying suitable directions for the proposed research work.

3.3.2 Review Methodology

The systematic review adopted the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework to identify, screen, and select relevant publications related to machine learning applications in mental healthcare. Studies were collected from multiple scientific databases using predefined search strategies and were screened according to inclusion and exclusion criteria to ensure the quality and relevance of the selected literature.

The reviewed studies were subsequently categorized according to the nature of the data employed and their application domains. This categorization enabled a systematic comparison of existing methodologies and highlighted the diversity of machine learning approaches applied to mental healthcare problems. During the reporting period, particular attention was given to understanding the contributions of the reviewed studies rather than only the review methodology itself.

3.3.3 Contributions of the Reviewed Studies

The systematic review categorized the selected studies according to the primary source of information used for mental health assessment. During the reporting period, representative papers from each category were studied to understand their research objectives, data sources, machine learning techniques, and overall contributions. The reviewed studies demonstrated that each category provides unique insights into mental health assessment while collectively advancing the development of intelligent clinical support systems.

Based on the organization adopted in the systematic review, the analysed studies are presented under the following categories:

- Sensor-Based Studies
- Text-Based Studies
- Structured Data Studies
- Multimodal Studies

Each category is discussed separately in the following subsections to highlight the characteristics, contributions, and limitations of the corresponding research works.

Sensor-Based Studies

Sensor-based studies utilize information collected from wearable devices, smartphones, and physiological sensors to continuously monitor behavioural and physiological characteristics associated with mental health conditions. Common data sources include accelerometers, GPS sensors, heart rate monitors, sleep trackers, and smartphone usage logs.

The reviewed studies demonstrated that these objective measurements can be analysed using machine learning algorithms to identify behavioural patterns related to depression, anxiety, stress, and emotional well-being. Their primary contribution lies in enabling continuous and unobtrusive monitoring outside traditional clinical environments, thereby supporting early detection and long-term monitoring of patients. These studies also highlighted the growing importance of wearable technologies in modern mental healthcare.

Text-Based Studies

Text-based studies investigate mental health by analysing linguistic information obtained from social media platforms, online discussion forums, clinical interviews, and electronic health records. These studies primarily employ natural language processing techniques to extract meaningful linguistic and semantic features from textual data.

The reviewed papers demonstrated that language usage, sentiment, writing style, and semantic patterns can effectively indicate symptoms of depression, suicidal ideation, anxiety, and other psychological disorders. Their major contribution is the development of scalable approaches for mental health screening using readily available textual information while emphasizing the importance of language as a behavioural marker for psychological assessment.

Structured Data Studies

Structured data studies utilize organized clinical information, including demographic characteristics, psychological questionnaires, medical histories, laboratory results, and electronic health records. These well-defined variables provide reliable input for statistical analysis and supervised machine learning models.

The reviewed studies primarily focused on diagnosis, patient classification, risk prediction, and treatment outcome assessment. Their contributions demonstrated that structured clinical data remain highly valuable for developing interpretable machine learning models capable of supporting evidence-based clinical decision making while complementing traditional psychological assessment procedures.

Multimodal Studies

Multimodal studies integrate multiple sources of information within a unified machine learning framework to obtain a more comprehensive understanding of an individual's mental health condition. The reviewed studies combined various data modalities, including physiological signals, wearable sensor measurements, textual information, speech, facial expressions, behavioural observations, and clinical records.

The analysis demonstrated that multimodal learning generally achieves higher predictive performance than single-modality approaches by exploiting complementary information from heterogeneous data sources. At the same time, these studies identified important challenges related to data integration, computational complexity, interpretability, privacy preservation, and deployment in real-world clinical settings, thereby highlighting promising directions for future research.

3.3.4 Comparative Analysis

The reviewed studies demonstrated that machine learning has been successfully applied to mental healthcare using diverse data sources and analytical methodologies. Although each category addressed different aspects of mental health assessment, they collectively contributed towards improving early diagnosis, continuous monitoring, symptom prediction, and clinical decision support.

Sensor-based studies emphasized continuous and objective monitoring of behavioural and physiological changes, enabling real-time assessment of patients outside traditional clinical environments. Text-based studies exploited linguistic and semantic information to detect psychological conditions, demonstrating the effectiveness of natural language processing for large-scale mental health screening. Structured data studies focused on clinically organized information, producing interpretable prediction models that complement conventional diagnostic procedures. In contrast, multimodal studies

integrated multiple sources of information to improve predictive performance by combining complementary behavioural, physiological, and clinical characteristics.

Overall, the comparative analysis revealed that no single data source is sufficient for comprehensive mental health assessment. Instead, each modality contributes unique information, and their integration offers greater potential for developing accurate, reliable, and clinically useful machine learning systems.

3.3.5 Research Gaps and Future Directions

Despite the significant progress reported in the reviewed studies, several research challenges remain. Many studies relied on relatively small or homogeneous datasets, limiting the generalizability of the developed models. In addition, the lack of standardized datasets and evaluation protocols makes it difficult to compare the performance of different machine learning approaches.

Another important limitation concerns the interpretability of machine learning models. While advanced learning algorithms often achieve high predictive accuracy, many operate as black-box models, reducing clinicians' confidence in their predictions. Privacy and ethical concerns associated with collecting personal behavioural and health-related data also continue to present significant barriers to real-world deployment.

Although multimodal learning demonstrated promising improvements in predictive performance, relatively few studies investigated comprehensive frameworks capable of integrating heterogeneous data while maintaining computational efficiency and model transparency. Furthermore, longitudinal studies involving long-term monitoring and personalized interventions remain limited.

These observations indicate that future research should focus on developing interpretable, privacy-preserving, and clinically deployable machine learning systems capable of integrating multiple complementary data sources while supporting personalized mental healthcare.

3.3.6 Learning Outcomes

The systematic review provided a comprehensive understanding of contemporary machine learning applications in mental healthcare from both technical and Human–Computer Interaction perspectives. Particular emphasis was placed on analysing the objectives, datasets, machine learning methodologies, and contributions of individual studies rather than only the overall findings of the review.

The literature survey also highlighted the strengths and limitations of existing research approaches, enabling a better understanding of the practical challenges associated with developing clinically applicable machine learning systems. The knowledge acquired through studying the reviewed papers played an important role in identifying suitable

datasets, understanding current research trends, and defining the direction of the proposed research work.

4. Conclusion

This report presented the work carried out during the reporting period to establish the statistical and methodological foundation required for the proposed research in machine learning applications for mental healthcare. The study combined theoretical understanding of statistical methodologies with a comprehensive analysis of existing literature, thereby providing the necessary background for future research activities.

The statistical component of the study covered both parametric and non-parametric hypothesis testing, including their mathematical formulations, assumptions, statistical power analysis, and sample size estimation procedures. The comparative analysis between corresponding parametric and non-parametric methods further strengthened the understanding of selecting appropriate statistical techniques under different experimental conditions and data characteristics.

The literature review complemented the statistical study by examining psychometric scale development and validation studies together with recent advances in machine learning for mental healthcare. The analysis of SERATS, SERAPTS, and FEATS provided practical insight into the statistical procedures employed in developing and validating psychological assessment instruments. Furthermore, the systematic review and the analysis of representative research papers highlighted the diverse applications of machine learning, the variety of data sources employed, and the strengths and limitations of existing methodologies.

Collectively, the studies presented in this report establish a comprehensive foundation in statistical analysis, psychometric evaluation, and machine learning methodologies. The knowledge acquired through these investigations provides a clear understanding of current research trends, existing challenges, and potential research opportunities, thereby supporting the development and implementation of the proposed research in a systematic and scientifically rigorous manner.

Bibliography

[1] Sarah Graham, Kathryn Depp, et al. Machine learning in mental health: A systematic review of the hci literature to support the development of effective and implementable ml systems. *ACM Transactions on Computer-Human Interaction*, 29(5), 2022. doi: 10.1145/3544548.

[2] S. Haeyen, J. van Hooren, and M. Hutschemaekers. Measuring the contribution of art therapy in multidisciplinary treatment of personality disorders: The construction and psychometric evaluation of the self-expression and emotion regulation in art therapy scale (serats). *Personality and Mental Health*, 12(2): 99–114, 2018. doi: 10.1002/pmh.1407.

[3] S. Haeyen, J. van Hooren, et al. Development and validation of the self-expression and emotion regulation in arts and psychomotor therapies scale (serapts). *The Arts in Psychotherapy*, 2025.

[4] Richard A. Johnson and Dean W. Wichern. *Applied Multivariate Statistical Analysis*. Pearson, 6 edition, 2007.

[5] E. L. Lehmann. *Nonparametrics: Statistical Methods Based on Ranks*. Springer, 2006.

[6] N. M. Ramachandran and Chris P. Tsokos. *Mathematical Statistics with Applications*. Academic Press, 2 edition, 2009.

[7] S. Snir and R. Regev. Validation of the formal elements art therapy scale: The problem solving scale. *The Arts in Psychotherapy*, 41(4):432–439, 2014.